Particles from Native Overlap Transport: Common Currents, Defect Modes and a Pixel-Dependent Mass Operator

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Abstract

Observer Patch Holography proposes that particles are reproducible features of the records of a specified observer, repair and memory process. We construct a finite link between that proposal's gauge and particle sectors: the same localized fermion modes carry the twelve native current operators and a pixel-dependent charged-lepton mass matrix. A complete memory process realizes the matrix with a controlled weak-step limit. Its feedback into a declared electromagnetic kernel has certified conditional fixed points; a minimal field census with one specified scale law is instead inconsistent. A compact opposite-winding texture has finite excess-energy bounds and four isolated modes with controlled infinite-volume errors. Its same-band quartic extension gives exact interacting spectral readouts; local guarded repair preserves both charges and admits controlled transport of the band. The compact splitting changes the interacting ground sector and its poles. The finite gauge theorems constrain the internal representation; they do not identify physical particles or fix all couplings. The resulting construction is a precise conditional particle model, without a complete observed spectrum, an interacting chiral continuum, or a common Einstein limit.

1 One source process and its particle interpretation

Observer Patch Holography (OPH) proposes a self-referential complete history: observers infer the specification of their world from shared records and realize it through operations within that same history. The executable version specifies a carrier Hilbert space, accessible algebras, repair instruments, a scheduler, memory, feedback and preparation services [2, 6]. These are permitted design choices. Their implementation and their observable consequences must agree; selecting a design is not itself a proof that it describes nature.

Operational agreement concerns accepted shared data. State selection and endogenous implementation of response transformations are additional stated principles, not consequences of self-reference alone. The exact basis and conditional theorem premises are recorded in the common registry [20]. This paper consumes the finite carrier and complete-response results of the gauge construction [5]; it states separately the spatial, dynamical and physical assumptions needed for matter.

The source specification is a tuple

$$\mathcal{S} = (\mathcal{H}, \mathcal{A}, \{\Phi_{e,r}\}, \sigma, \mathcal{M}, \mathcal{F}, \mathcal{P}),$$

where $\Phi_{e,r}$ is the operation on seam e with retained outcome r, σ selects operations, $\mathcal{M}$ is memory, $\mathcal{F}$ is feedback, and $\mathcal{P}$ is the preparation domain. A particle assertion concerns the joint and conditional laws of actual preparations, interventions and detector outputs of $\mathcal{S}$. Agreement between two spectra alone does not identify the states, currents, clocks or instruments that produced them.

Hypothesis 1.1 (Matter as localized defects)**.** *Stable massive matter admits a localized excitation of a continuing source reference, with an invariant preserved by the admitted dynamics. Its creation, transport, interactions and detection are operations of the same source. Its total excess energy and propagation use one energy and clock dictionary. Unstable species may be metastable or internal excitations; massless radiation need not carry a defect invariant.*

This hypothesis does not require a distinct conserved topological charge for every mass eigenstate. Flavor mixing and decay can connect internal states of a defect family. A protected label cannot change at an admitted vertex without another field or record carrying the compensating charge. A proposed complete particle catalogue must account for those vertices as well as masses.

The construction developed here has the following dependency structure:

$$\begin{array}{ccccc} \text{native currents} & \longrightarrow & M_{15} \otimes \mathbb{C}^3 & \longrightarrow & \text{Higgs invariant lines} \\ P & \longrightarrow & t_i(P) & \longrightarrow & g(P) Z_C(P) \\ \text{texture } h & \longrightarrow & W & \longrightarrow & E^\dagger H E,\ E^\dagger J E. \end{array}$$

The first row constrains internal quantum numbers, the second constructs a chosen mass operator, and the third attaches both to common localized states. Closing the physical model additionally requires that the same full interacting current response determine P, rather than inserting a separate response model. The finite statements below make this distinction explicit.

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2 What the finite gauge theorems determine

2.1 Complete response and the compact Lie algebra

The carrier is the oriented icosahedral boundary, with twelve ports, thirty seams and twenty faces. Its real port module under the proper rotation group A_5 decomposes as

$$V_{12} = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}, \qquad \dim V_{12}^{A_5} = 1.$$

The following is the finite input to the particle construction. Assume a faithful real-linear response map $D : V_{12} \to \mathfrak{u}(\mathcal{K})$ whose image is the complete connected response algebra and is closed under commutators. Assume proper carrier rotations act on that image through the same connected inner response group, with operationally silent centralizer factors allowed. Then the compact classification gives

$$\mathfrak{g} \simeq \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3). \tag{1}$$

The short dimension argument explains the strength of the assumptions. A compact Lie algebra is the sum of its centre and its semisimple part. Inner rotations fix the centre, so the one-dimensional invariant subspace bounds its dimension by one. A twelve-dimensional centre-free compact algebra would be $\mathfrak{su}(2)^4$ under the small-dimension classification. Inner A_5 actions preserve each factor; the fixed subspace in each three-dimensional factor has dimension zero or three. Their sum cannot have fixed dimension one. The centre therefore has dimension one, leaving the unique semisimple dimension split $11 = 3 + 8$ [5].

Twelve ports alone do not imply (1). Dropping completeness, bracket closure, faithfulness or the inner-action condition changes the problem. In particular, a diagonal port response can be abelian, and allowing further connected directions can produce a larger Lie algebra. The finite result also does not identify carrier rechartings with physical spatial rotations of a relativistic scattering theory.

There is an explicit chosen current law on the protected native carrier. For port vectors v_p, their Galois-conjugate triplet $\sigma(v_p)$ and the cross-product matrix $\widehat{v}$, set

$$D_p = \operatorname{diag}\left(\widehat{v_p/2} + iQ_p,\ \widehat{\sigma(v_p)/2}\right), \qquad Q_p = \tfrac{1}{2} v_p v_p^T - \tfrac{1}{6}\|v_p\|^2 I_3 + \tfrac{1}{12} I_3.$$

The twelve matrices have rank twelve as response directions; their brackets close, and the actual port permutations implement the required covariance. Their sum is $\operatorname{diag}(iI_3, 0)$, an observable relative phase under cross-triplet readout. This realizes $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$ and hence the stated Lie type. Coherent control, encoding and record-assisted recovery belong to this specified law; unchanged dissipative C/E repair alone is not assigned these reversible generators [5, 6].

2.2 The matter module and its family factor

A declared spin lift to five internal coordinates and the nonvacuum even exterior module give

$$M_{15} = \Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5, \qquad \dim M_{15} = 10 + 5 = 15.$$

The actual source-current lift preserves all brackets and has an exact intertwiner with this module. Its simultaneous primitive-charge and color/weak Casimir readouts give the following all-left convention.

Label	Color	Weak	$6Y$	Components
Q	$\mathbf{3}$	$\mathbf{2}$	1	6
u^c	$\overline{\mathbf{3}}$	$\mathbf{1}$	-4	3
d^c	$\overline{\mathbf{3}}$	$\mathbf{1}$	2	3
L	$\mathbf{1}$	$\mathbf{2}$	-3	2
e^c	$\mathbf{1}$	$\mathbf{1}$	6	1

The labels abbreviate computed finite quantum numbers. Their identification with laboratory Weyl fields requires the physical current, spin and kinetic maps. The selected exterior grammar has two conjugate anomaly-free masks; conjugating the mask does not add a generation. The exact identities include

$$\mathrm{tr}(6Y) = \mathrm{tr}[(6Y)^3] = 0, \quad 2(1) - 4 + 2 = 0, \quad 3(1) - 3 = 0,$$

for the abelian and mixed anomalies, with $2 - 1 - 1 = 0$ for the color cubic anomaly and four weak doublets per copy for weak global parity. These are representation statements, conditional on the intended chiral field interpretation.

The maximal faithful image of these local representations has common central kernel $\mathbb{Z}_6$. Selecting a physical global gauge group and allowed line operators needs the corresponding source loop and line data. The finite kernel computation alone does not exclude a different covering-group interpretation with the same local charges.

The finite family theorem selects a rank-three band under a complete faithful band premise, the stated dimension window and strict seam-cost comparison. Applying it to particles requires an isometry from that same band into the family factor. In the executable field construction we explicitly choose $M_{15} \otimes \mathbb{C}^3$, with gauge currents acting trivially on $\mathbb{C}^3$. The family choice is allowed simulator design. A two-dimensional radial kernel or a winding number three is not a proof of three generations. Equivalent triplet representations on spatial, color and family factors do not identify those factors.

There is a separate finite route to protected exchange records. On the native ordered-cover transport, let $p_h(x)$ be the permutation parity accumulated by branch h. A coherent ordering bit gives

$$K_h = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes X^{p_h(x)}, \qquad Q = I \otimes X, \qquad [K_h, Q] = 0.$$

An odd exchange loop acts as X on the bit, and the corresponding colored fermionic lift acts as $-$SWAP. The sectors survive complete record-adaptive native schedules in this specified memory design. Dephasing or resetting the bit changes that conclusion. These are exchange-history sectors, distinct from spatially localized texture charge; they do not establish the physical spin and statistics of the bound modes constructed below. The source constructions are in `model/defect_protection` and `model/charged_defect`.

3 The conditional field action and particle table

3.1 Symmetry constrains the action, not its numerical coefficients

Choose one complex weak doublet H of hypercharge $1/2$, canonical positive kinetic forms, physical Weyl statistics and a Lorentz-covariant local field interpretation in $3+1$ dimensions. Restrict the polynomial action to operator dimension at most four, modulo integration by parts. The invariant grammar then gives the gauge kinetic terms, scalar kinetic term and potential, fermion kinetic terms and exactly three charged Yukawa lines. Independent gauge couplings, topological theta terms and an additive vacuum-energy constant also remain free; local integration by parts does not remove their possible global physical effects. In the four-component convention the charged Yukawa part is

$$\mathcal{L}_Y = -\overline{Q}_L \widetilde{H} Y_u u_R - \overline{Q}_L H Y_d d_R - \overline{L}_L H Y_e e_R + \text{h.c.}$$

Each Y_a is a complex 3×3 matrix. The all-left source convention uses the corresponding conjugated blocks; changing only half the convention would change the charge equations. The native invariant-line computation solves all twelve current equations for both Higgs components and gives one line for each allowed origin $(6Y_1, 6Y_2) = (-4,1), (1,2), (-3,6)$. Tensoring these lines with the gauge-trivial family factor leaves exactly the three matrix coefficients. Canonical kinetic normalization must precede any reading of their singular values as action masses [5, 19, 18].

For the explicitly chosen broken potential

$$V(H) = \lambda_H (H^\dagger H - v^2/2)^2, \qquad \lambda_H > 0,$$

the vacuum shell has $|H| = v/\sqrt{2}$. Taking $H = (0, v/\sqrt{2})$ gives the following conditional quadratic table.

Field directions	Action mass or coefficient
Three charged leptons	$m_{e,i} = v\, s_i(Y_e)/\sqrt{2}$
Three up and three down flavors	$m_{a,i} = v\, s_i(Y_a)/\sqrt{2}$, each in three colors
Three minimal left neutrino fields	Zero in the dimension-four charged grammar
Charged weak vectors	$m_W = g_2 v/2$
Neutral massive weak vector	$m_Z = v\sqrt{g_2^2 + g_Y^2}/2$
Electromagnetic direction	Zero quadratic mass
Eight color directions	Zero quadratic mass; no free physical gluon claim
One radial scalar	$m_h = \sqrt{2\lambda_H}\, v$

The neutral vector matrix follows from the coefficient of $\frac{1}{2}A^T M^2 A$ in $|DH|^2$; this factor of two matters. The three zero scalar Hessian directions are gauge-orbit directions and do not count as three additional physical scalars. Neither the sign of the broken potential nor its quantum vacuum is selected by the finite representation. This action and its normalization are checked in `model/conditional_standard_model`.

For a supplied Weyl principal symbol and the antisymmetric pairing $\epsilon_{\text{spin}} \otimes \overline{K}$, squaring the complete Nambu symbol cancels its momentum–mass cross terms. Its frequencies obey

$$E_j(\mathbf{p})^2 = c^2|\mathbf{p}|^2 + s_j(K)^2.$$

This is a theorem of that supplied relativistic action. The spatial source constructed below has additive bands instead, so the displayed relativistic formula cannot be transferred to it by matching K alone.

3.2 Flavor, neutrinos and conserved defect labels

The canonical family-frame freedom is $\mathrm{U}(3)^5$. It preserves the charged singular spectra and basis-invariant mixing data. In particular $V_{\mathrm{CKM}} = U_{uL}^{\dagger} U_{dL}$ is determined by the same left-handed weak current; relabeling families cannot repair a mass hierarchy or generate physical CP violation. A real symmetric coefficient class has $\mathrm{tr}[Y_u Y_u^{\dagger}, Y_d Y_d^{\dagger}]^3 = 0$. This is a restriction of that class, not of arbitrary complex source feedback [17].

A supplied dimension-five operator $(L \cdot H)^T C_5 (L \cdot H)/(2\Lambda) + \mathrm{h.c.}$ gives $M_\nu = v^2 C_5/(2\Lambda)$ and Takagi singular masses. It supplies neither C_5 nor its scale from the charged-lepton construction. The source checks are in `model/majorana_defect_bridge`.

Conserved additive defect labels must respect the same vertices. If the charged-current mixing graph is connected, the relation

$$q(W^-) - q(e_\alpha) + q(\nu_i) = 0$$

for every nonzero entry forces one common neutrino label N and one common charged-lepton label. A massive Majorana field adds $2N = 0$. In an integer winding group this implies $N = 0$. If a massless neutrino is excluded from defect assignments and has label zero, connected mixing forces the other neutrino labels to zero as well. Distinct protected winding numbers therefore cannot simply be assigned to the three mixed mass eigenstates. Internal excitations, nonadditive invariants and charged compensating vertices require their own source construction.

4 Pixel capacity and the chosen mass law

The pixel variable is the dimensionless capacity ratio $P = a_{\mathrm{cell}}/\ell_P^2$. It is distinct from a projector P_j in the memory controller. The declared self-reference relation reads

$$P = \varphi + \sqrt{\pi}\,\alpha, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

The inner response supplying α must be specified. In the heat/running construction it depends on P through a unified scale, a heat-sum equation, low-energy couplings and charged masses. Thus the loop has the form

$$P \longrightarrow (\alpha_U, v, \text{field census}) \longrightarrow (M_f, J_f) \longrightarrow A_{\mathrm{response}} \longrightarrow \varphi + \frac{\sqrt{\pi}}{A_{\mathrm{response}}}.$$

The same masses and current normalization must appear on both sides. A fixed point of a declared response proves consistency of that map, without identifying it as the physical electromagnetic response.

The mass law used here is a chosen finite response architecture with icosahedral incidence registers and a determinant normalization. Its charged shape has a three-phase form related to Koide-type parametrizations [7, 8]. The architecture has target-informed provenance; its predictions are conditional postdictions even when its execution reads no measured masses. Choosing it as part of a simulator is allowed. Independent validation requires consequences not used to choose it. The following realization states both the operator identity and the complete-record error bound. Its displayed $\beta = 4$ scale convention is one specified branch; the normalized coefficients and memory identities also apply to the explicit $\beta = 7$ branch considered afterward.

4.1 A pixel mass law in native memory dynamics

A chosen simulator specification can realize the conditional pixel-dependent charged mass law on the native gauge-family carrier. The construction uses the twelve-port reset controller and the complete native current instruments. Its design inputs are independent left and right probe durations, a rule enabling feedback on a successful C event, and a common feedback amplitude. The conditional mass-law architecture is target-informed; evaluating that architecture without a charged-mass input does not establish a prospective prediction.

The selected response architecture gives

$$\kappa = \frac{\alpha_U/50}{1+\alpha_U/31+\alpha_U^2/310}, \qquad \chi = \frac{\alpha_U^2/512}{1-\alpha_U/77}, \tag{2}$$

$$\zeta = \frac{(P-\varphi)^2/(21\pi)}{1-\alpha_U/27-\alpha_U^2/135}, \qquad r_j = 1+\sqrt{2}e^{-\chi}\cos\left(\frac{2}{9}+\zeta+\frac{2\pi j}{3}\right), \tag{3}$$

where $\varphi = (1+\sqrt{5})/2$ and $j = 0,1,2$. Its charged coefficients, ordered increasingly, are

$$t_i = \frac{m_i}{v} = \frac{e^{\kappa} r_i^2}{\left(2\,6^{14}\prod_{j=0}^{2} r_j^2\right)^{1/3}}, \qquad \frac{v}{E_*} = P^{-1/2}\exp\left(-\frac{2\pi}{4\alpha_U}\right). \tag{4}$$

The register dimensions and determinant exponent in these formulas are part of the selected architecture. The scale E_* is a source coordinate; assigning it an energy in laboratory units requires a clock calibration.

Let $t_i > 0$ be these three charged coefficients and $\lambda_i = t_i/\sum_j t_j$. For the native projectors P_2, P_3, P_{10}, direct multiplication gives $\operatorname{tr}(P_iP_j) = 1/5$ for distinct indices and the spanning-vector Gram determinant $d = 2/5 - 2\sqrt{5}/25$. Consequently the matrix $Y = uP_2 + v_1P_3 + wP_{10}$ has elementary eigenvalue invariants

$$\operatorname{tr} Y = u + v_1 + w, \qquad e_2(Y) = \frac{4}{5}(uv_1 + uw + v_1w), \qquad \det Y = duv_1w. \tag{5}$$

Choose its three positive weights as the roots of

$$z^3 - z^2 + \frac{5}{4}e_2(\lambda)z - \frac{e_3(\lambda)}{d} = 0, \qquad w < u < v_1. \tag{6}$$

It follows that $\operatorname{spec} Y = \{\lambda_i\}$.

The complete C-enabled memory coefficient, with $a = p(1-q)$, $b = pq$ and $D = 1 - a^2 + ab$, is

$$Z_C = \frac{(1-a)P_2 + a(1-a)P_3 + abP_{10}}{D} - (1-p)P_2. \tag{7}$$

It has trace p; disabled records have not been removed or normalized away. Put $A = v_1(v_1+w)$ and $B = v_1(1-v_1+w) + (v_1+w)w$. The inverse equations

$$-Ap^2 + Bp - w = 0, \qquad q = 1 - \frac{v_1}{1-p(v_1+w)} \tag{8}$$

give the exact identity $Z_C = pY$. This follows by substituting the second equation and reducing each of the three coefficient differences by the quadratic in the first equation.

At each of the supplied forward and measured-endpoint pixel coordinates, rational Taylor enclosures and polynomial endpoint signs give exactly three roots of (6), and exactly two solutions of (8)

satisfying $0 < p < 1/12$, $0 < q < 1/6$. For the forward coordinate the first solution is approximately $p = 0.01915526454$, $q = 0.05474497127$. The native charged-lepton support Casimirs are $9/8$ and 3, so the complete probes implement these rates with the declared durations

$$\theta_L = \frac{\arcsin\sqrt{12p}}{\sqrt{9/8}}, \qquad \theta_R = \frac{\arcsin\sqrt{6q}}{\sqrt{3}}. \tag{9}$$

Both probes retain all thirteen outcomes. Their product retains all 169×128 current and repair labels per old memory input. The covariant binary grouping retains 512 labels per input and all twelve old inputs.

Choose $g = \sum_i t_i/p$. Then gZ_C has exactly the three coefficients t_i. Its diagonalizing unitary acts on the gauge-trivial family factor, commutes with every native gauge current, and preserves the canonical kinetic form. Thus this is an operator realization of the conditional mass law, up to the allowed family frame. It does not assert an additional alternating-group intertwiner. The source-coordinate enclosures also give $\sum_i t_i < 1$ by bounding the cube of that sum.

The coefficient is the generator of a controlled limit of the full correlated memory. Apply $\exp(-ihgX \otimes P_{\mathrm{new}})$ on enabled records and identity on disabled records. Let Ψ_h be this complete classical/quantum channel. Its classical transition satisfies $T = (1-p)R + pS$, where R resets to port two and S is stochastic. For every normalized initial classical/quantum state z_0, including an arbitrary untouched finite reference, the reduced state after N steps of size $h = \tau/N$ obeys

$$\left\| \sum_{\mathrm{memory}} \Psi_{\tau/N}^N(z_0) - \mathcal{U}_\tau\left(\sum_{\mathrm{memory}} z_0\right) \right\|_1 \le \frac{4g\tau}{N(1-p)} + \left(\frac{8}{1-p} + 4\right)\frac{g^2\tau^2}{N}, \tag{10}$$

where $\mathcal{U}_\tau$ is generated by $gX \otimes Z_C$. For a stationary product preparation the first term vanishes.

To prove the bound, let $\Pi z = \pi \sum z$ and $e_k = (I - \Pi)z_k$. The sum of trace norms on memory blocks gives $\|Te\|_1 \le p\|e\|_1$ on zero-sum blocks and $\|\Psi_h - T\| \le 2gh$. Hence $\|e_k\|_1 \le 2p^k + 4gh/(1-p)$. The reduced stationary step and the mean unitary step have the same first derivative; their two Taylor remainders sum to at most $4g^2h^2$. The reduced error due to e_k is at most $2gh\|e_k\|_1$. Summing these errors proves (10).

This limit is in dimensionless simulator time. A finite unread channel need not equal its limiting unitary. The feedback pulse service is an explicit design choice; compilation into a smaller native gate alphabet is a separate requirement. The forward pixel coordinate has no complete physical endpoint proof, and the measured-endpoint coordinate inherits measured-alpha calibration. No physical pole-mass identification, renormalization map, operational energy scale, or complete quark and neutrino spectrum follows from this finite realization.

4.2 Pixel feedback with one charged-fermion mass prescription

The pixel feedback must use the same mass coefficients as the declared field and memory operators. Let $u = \alpha_U$ and retain

$$M_U/E_* = e^{-2\pi}P^{1/6}, \qquad v/E_* = P^{-1/2}e^{-2\pi/(\beta u)}, \qquad \alpha_i^{-1} = u^{-1} + \frac{b_i}{2\pi}\log(M_U/m_Z).$$

Here E_* is a declared energy unit. The same conditional three-lepton coefficients are inserted into the charged-fermion integral, together with all six declared quark coefficients, including the finite top contribution. With the prescribed factor $1 - 3\alpha_3/\pi$ on the quark terms, the resulting inverse response $A(P)$ defines $P = \phi + \sqrt{\pi}/A(P)$. This kernel supplies no charged-vector or charged-scalar terms, confining hadronic spectral measure, threshold matching, or derivation of the native current's physical Thomson limit.

There is an exact obstruction to substituting the minimal Standard Model census while retaining $\beta = 4$. Assume $P \geq 1$, $M_U \geq m_Z$, positive couplings and threshold-free one-loop coefficients $b = (41/10, -19/6, -7)$. Set $x = \alpha_2^{-1}$ and $y = \alpha_3^{-1}$. Eliminating the unified coupling and scale ratio gives $0 < y \leq x$ and

$$\frac{\pi(18x + 5y)}{46} = 2\pi - \frac{2}{3}\log P + \frac{1}{2}\log\left[\pi\left(\frac{1}{x} + \frac{69}{333x - 218y}\right)\right].$$

The logarithm's argument is at most $8\pi/(5x)$. The function $9\pi x/23 - 2\pi - \frac{1}{2}\log[8\pi/(5x)]$ is increasing and exceeds one at $x = 6$, so $x < 6$ and both nonabelian couplings exceed $1/6$. Their heat-sum denominators are at least one; direct outward summation with infinite-tail bounds places their combined numerators below $1/80$. The required pixel entropy is $P/4 \geq 1/4$, a contradiction. This excludes the stated scale and census combination, independently of the readout kernel.

Two other declared maps have certified local fixed points. The supersymmetric running proxy with $\beta = 4$ gives $P \simeq 1.630971514159316$ and $A \simeq 137.000994766342$. The minimal census with the explicit scale choice $\beta = |b_3| = 7$ gives $P \simeq 1.628533369611833$ and $A \simeq 168.815082928453$. The first requires extra supersymmetric fields absent from the native quadratic carrier. The second supplies a scale choice, without identifying the Higgs scale physically with color transmutation. For each map, verified implicit-root signs, infinite heat tails and interval automatic differentiation give a strict self-map and $|g'| < 0.00734$ on the reported coupling interval. Uniqueness is local to that interval.

Rational interval inversion over both complete root enclosures admits both native memory solutions, with $0 < p < 1/12$ and $0 < q < 1/6$. The gain $g = \sum_j (m_j/v)/p$ preserves the unconditional duty probability, so the same full-record memory coefficient has the required three singular energies. The proof and independent simultaneous-root and quadrature checks are in `model/pixel_mass_closure`. These are conditional consistent feedback calculations; neither is a complete physical particle or fine-structure prediction.

The beta coefficients in the feedback calculation are obtained from the actual internal-current traces with the intended chiral field multiplicities. This assumes that those are the surviving continuum fields. It does not count or remove spatial regulator valleys or doublers. The minimal-census root and the supersymmetric proxy are different scale designs, not two calibrations of one proved physical theory. The absence of a positive solution for the minimal census with $\beta = 4$ is a mathematical inconsistency of that specified combination; it is not an exclusion of other admitted OPH designs.

The three increasing coefficients returned by these closed conditional maps are shown below. They are dimensionless singular coefficients m_i/v; there is no physical pole-mass or GeV assignment in this table.

Scale design	m_1/v	m_2/v	m_3/v
Supersymmetric proxy, $\beta = 4$	$2.07203042 \times 10^{-6}$	$4.28430204 \times 10^{-4}$	$7.20520710 \times 10^{-3}$
Minimal census, $\beta = 7$	$2.07103435 \times 10^{-6}$	$4.28253912 \times 10^{-4}$	$7.20232110 \times 10^{-3}$

The interval certificate encloses these values and both admissible memory inverses over each complete root box, so the table and the controller refer to the same conditional source coordinates.

5 Localizing the same current and mass operators

5.1 Topology, total energy and the radial source

The seed of the defect hypothesis is a precise finite frustration statement. For a finite connected graph, abelian edge offsets b_e and positive weights w_e, define

$$\Phi_b(x) = \sum_{e:u\to v} w_e \, \mathbf{1}[x_v - x_u \neq b_e].$$

If a cycle has nonzero signed offset sum, then $\min_x \Phi_b(x) > 0$. Indeed, the possible costs form a finite set of positive-weight subset sums; a zero cost would solve every edge equation and force zero cycle sums. This is frustration at fixed compatibility data [3]. It does not protect b against an admitted update that changes its class, localize its energy, or assign a rest mass.

For a one-cochain b on the full carrier complex, face curvature is $F = \delta b$ and $b(\partial c) = F(c)$ for every two-chain c. The icosahedral boundary has trivial first homology: graph cycles bound faces. The native load difference $b_{uv} = n_v - n_u$ is exact, so $F = 0$ and all loop sums vanish. An actual defect invariant requires additional transport or field data; it does not follow from nonuniform loads alone.

Consider the two-spin cubic hopping operator with forward link $i\sigma_j/2$ and its Hermitian reverse. A declared radial onsite texture is

$$B_x = M \sum_{j=1}^{3} (-1)^{x_j} \frac{x_j}{|x|} \sigma_j, \qquad B_0 = 0.$$

On the odd radius-one cube at $M = 64$, this literal source has exactly two zero modes and a complementary absolute-energy gap at least 62. The normalized kernel frame W has scalar local density $W_x^\dagger W_x = \rho_x I_2$, with $\sum_x \rho_x = 1$. The kernel and full complement are retained in the following construction; the finite spin coordinates are not identified with physical Lorentz spin by this fact alone.

A radial mass profile also needs a total energy law. In the selected ordered-spin rotation orbit, each admissible charged shell costs at least $\pi J \kappa |q|$ with $\kappa > 0$. A hedgehog retaining nonzero degree on arbitrarily many disjoint shells therefore has infinite excess energy in that ungauged identification. A fermion bound-mode gap omits this background cost. Screening, compact opposite-charge textures or another specified finite-energy structure are substantive dynamical choices. The compact pair below provides the latter type of finite source example.

5.2 One local carrier for defect states, currents and mass coefficients

The radial defect operator admits a direct realization of the native matter currents and a prescribed family mass operator on the same localized modes. The construction is a choice of simulator carrier and coherent pairing service. Its finite operator identity is exact; the choice is not claimed to follow uniquely from overlap consistency.

Let h be the radial operator on $\ell^2(\Lambda) \otimes \mathbb{C}^2$, with its normalized kernel frame $W : \mathbb{C}^2 \to \ker h$. Use the strong texture domain, in which the kernel has dimension two and the complementary spectrum has absolute energy at least $d > 0$. With

$$J = i\sigma_2, \qquad D_x = (-1)^{x_1+x_2+x_3}, \qquad C = DJ,$$

the source identities are

$$C^T = -C, \quad C^\dagger C = I, \quad hC = -C\overline{h}, \qquad hW = 0, \quad W^\dagger W = I_2, \quad C\overline{W} = WJ.$$

The last identity fixes the quaternion frame through evaluation at the origin and its scalar normalization Gram matrix.

Choose the canonical anticommutation relation algebra on $\ell^2(\Lambda) \otimes \mathbb{C}^2 \otimes M_{15} \otimes \mathbb{C}^3$. Here M_{15} is the native matter representation, transported by its actual source intertwiner; the three family copies are declared inputs. For a symmetric all-left Higgs kernel $K \in M_{45}(\mathbb{C})$, define

$$\Delta = C \otimes \overline{K}, \qquad B = \begin{pmatrix} h \otimes I & \Delta \\ \Delta^\dagger & -\overline{h} \otimes I \end{pmatrix}, \qquad E = \operatorname{diag}(W \otimes I, \overline{W} \otimes I).$$

The antisymmetry of Δ makes it a valid local creation pairing. Multiplication gives the exact isometric intertwiner

$$BE = EB_0, \qquad B_0 = \begin{pmatrix} 0 & J \otimes \overline{K} \\ (J \otimes \overline{K})^\dagger & 0 \end{pmatrix}. \tag{11}$$

Every singular value of K therefore appears with both signs and twice the multiplicity in the localized Nambu operator, including zero values. Nambu coordinates duplicate annihilation and creation operators; they introduce no physical mirror carrier.

For every actual anti-Hermitian native current G, the map $W \otimes I$ intertwines $I \otimes G$ with $I_2 \otimes G$. In Nambu coordinates the generator is $\operatorname{diag}(G, \overline{G})$; Hermitian charges use $\operatorname{diag}(Q, -\overline{Q})$. The Higgs background transforms with matter under the uniform internal gauge action. A fixed broken vacuum does not make every gauge generator commute with its mass operator. The resulting map of canonical anticommutation relation algebras intertwines quadratic Heisenberg evolution. On even states, tensoring with the normalized trace on the orthogonal complement gives an exact state inclusion with partial-trace recovery. This complement state is not a vacuum.

Use the positive family matrix $M(P)$ supplied by the memory construction, whose eigenvalues are the retained charged-lepton coefficients $m_i(P)/v(P)$. Choosing $H = (0, v(P)/\sqrt{2})$ and $Y_e = \sqrt{2}M(P)$ realizes those coefficients in $K/v(P)$ itself. The coherent pairing service implements the stationary-mean memory coefficient times its declared gain. Equation (11) does not identify the unread finite memory channel with this Hamiltonian. The calibration provenance of the retained family law is unchanged.

An explicit record service realizes the same Hamiltonian in a controlled weak-step limit. Apply the native current probes to fresh auxiliary legs in their invariant preparations, so their complete twelve-state memory transition is independent of the spatial field. For the updated memory port j choose the field pulse generated by

$$H_r = H_0 + \mathbf{1}_C(r)\, gL(P_j),$$

where H_0 contains the radial kinetic term and the chosen quark pairing, L is the local lepton pairing map, and g is the source gain. Every H_r has the same core intertwiner. Linearity gives stationary mean $\overline{H} = H_0 + L(gZ_C)$, precisely the Hamiltonian above. If $\|H_r\| \le M$ on the fixed finite Fock space, the reset contraction of the complete memory law and a second-order unitary expansion give

$$\left\| \rho_N - e^{-iT\overline{H}} \rho_0 e^{iT\overline{H}} \right\|_1 \le \frac{4MT}{N(1-p)} + \left(\frac{8}{1-p} + 4 \right) \frac{M^2T^2}{N}.$$

The bound includes arbitrary finite untouched references and initial memory correlations; stationary product memory removes the first term. For the displayed cube, $M = 2430 \cdot 66 + 1215(1+g)$ suffices. This chosen coherent service uses every record. Its finite unread channel need not be unitary, and its bound is not uniform in volume.

For a complete finite witness, choose the retained quark design $Y_u = \mathrm{diag}(6^{-6}, 6^{-3}, 1)$ and $Y_d = \mathrm{diag}(6^{-6}, 6^{-4}, 6^{-2})$, and set the three minimal neutrino coefficients to zero. Their exponents are design inputs, not topological index conclusions. On the radius-one cube there are 27 sites, 2430 physical annihilator modes and 4860 Nambu rows. The localized sector has 90 annihilators and 180 Nambu rows: 84 positive, 84 negative and 12 zero rows. These numbers count operator degrees of freedom, not observed species. In particular the zero rows do not establish a relativistic massless neutrino sector.

The full complement is retained. Indeed, with $T = \mathrm{diag}(I, C^\dagger \otimes I)$,

$$T^\dagger B T = \begin{pmatrix} h \otimes I & I \otimes \overline{K} \\ I \otimes K & h \otimes I \end{pmatrix}.$$

Its energies are the sums $\lambda(h) \pm s_j(K)$, not a relativistic square-root dispersion. If $\|K\| \le M$ and $2M < d$, the localized band lies in $[-M, M]$ and the complement has absolute energy at least $d - M$. This is single-quasiparticle isolation, not isolation of all many-body energy sums. In the displayed design, choosing $v(P)$ as energy unit, radial hopping one and texture coefficient 64 admits $d = 62$ and $M = 1$.

The common spatial density follows from $W_x^\dagger W_x = \rho_x I_2$; inversion gives mean position zero. A Fourier transform supplies a momentum distribution, without making the localized state a momentum eigenstate. Occupation of these modes does not create the supplied texture winding. Original site-resolved repair can leak out of the kernel, and interactions can mix the two factors. Thus the exact common finite operator supplies neither defect formation, physical rest masses nor an interacting relativistic continuum.

The package `model/defect_current_bridge/` retains all twelve native current operators. Its complete finite controls include an unstaggered-pairing counterexample.

5.3 Compact opposite-degree textures and their isolated modes

A finite-energy texture can carry opposite local winding numbers while approaching a constant exterior. On the two-spin cubic hopping carrier, choose an integer $a \ge 1$, write $R = a + 1$, and define

$$f(x, y, z) = \begin{cases} (x, y, z^2 - a^2), & \max(|x|, |y|, |z|) \le R, \\ (0, 0, 1), & \text{otherwise.} \end{cases}$$

Set $n = f/|f|$ away from $c_\pm = (0, 0, \pm a)$ and $n(c_\pm) = 0$. The Hamiltonian is $h = T + Mn \cdot \sigma$, where the nearest-neighbor hopping has $\|T\| \le \sqrt{3} < 2$ and $M \ge 8$. The texture and its amplitude zeros are explicit design inputs. The complete unit cubic shells about c_+ and c_- have degrees $+1$ and -1 under their normalized barycentric interpolation; the exterior shell is constant. Exact rational regular-value counts check every oriented triangle. This degree classifies the interpolated texture and is not a fermionic superselection charge.

For the selected ordered-spin reference, fill the two core directions by the exterior direction. Each rotated-ground source bond has excess energy in $[0, J]$. With $s = 2a + 3$, precisely $3s^2(s + 1)$ bonds touch the support cube, so

$$0 \le \Delta E_{\mathrm{spin}} \le 3s^2(s + 1)J.$$

The two zero fermion masses and the spin-core fill are distinct declared parts of the construction. The second-quantized Hamiltonian differs from that of the uniform texture by an operator of norm at most $2M(s^3 - 1)$. Its vacuum-energy difference has the same bound, independently of exterior

volume. These are total texture-energy bounds; they do not assign a constituent rest mass to a localized mode.

Projection onto the two core sites gives $h = \left(\begin{smallmatrix} 0 & B^\dagger \\ B & D \end{smallmatrix}\right)$. The exterior block has no spectrum in $(-M+2, M-2)$, and $\|B\|^2 = b^2$, where $b^2 = 7/4$ for $a = 1$ and $b^2 = 3/2$ otherwise. There are exactly four low eigenvalues. Their Feshbach equations give

$$|E| \le r := \frac{b^2}{M-4}, \qquad \frac{\|\psi_{\text{exterior}}\|^2}{\|\psi_{\text{cores}}\|^2} \le \frac{b^2}{(M-2-r)^2}.$$

These bounds are uniform in volume. For a cube of radius $L \ge R+1$, resolvent path expansion and monotone Feshbach eigenvalue crossings give

$$|E_j(L) - E_j(\infty)| \le \frac{2b^2}{M-r-2}\left(\frac{2}{M-r}\right)^{2(L+1-R)}.$$

Thus the infinite lattice also has precisely four isolated modes in $[-2, 2]$. At $a = 1$, $M = 64$, $L = 3$, complete rational interval certification of all 686 eigenvalues excludes zero from all four low levels and from their infinite-volume limits. The numerical levels are approximately ± 0.003905899328 and ± 0.003906437151. The finite-volume filled-sea energy difference is approximately 127.9700 in the same hopping units.

Let W span this four-dimensional band, $A = W^\dagger h W$, and $C = D_{\text{stagger}} i\sigma_2$. Particle-hole symmetry gives $C\overline{W} = WS$ with $S = W^\dagger C \overline{W}$ antisymmetric and unitary. The actual twelve native current operators intertwine through $W \otimes I_{45}$. The same local pairing $\Delta = C \otimes \overline{K}$ has the exact core

$$\begin{pmatrix} A \otimes I & S \otimes \overline{K} \\ (S \otimes \overline{K})^\dagger & -\overline{A} \otimes I \end{pmatrix}.$$

Its energies are those of A plus or minus the singular values of K. Consequently the compact texture shifts the retained family coefficients; dropping A gives an incorrect operator inclusion. A local Pauli perturbation at the two cores preserves every shell datum while changing these energies. Winding alone therefore does not pin a zero mode for this carrier. The construction supplies a compact background and controlled localized spectrum, without autonomous formation, stability, a relativistic mass shell or physical particle identification. The complete calculation and exact certificates are in `model/compact_defect_pair`.

5.4 Protected texture records and controlled band transport

A declared local repair law protects the compact pair's two winding charges. Let f_x be its unnormalized reference texture and fix the zero cores and constant exterior. At every noncore site retain a rational vector v_x with $\|v_x - f_x\| \le \eta \|f_x\|$, $0 < \eta < 1/4$, using $v_x/\|v_x\|$ in the actual Pauli potential. A one-vertex proposal is accepted only if it satisfies this budget and has nonnegative scalar product with the old vector and every other vertex of every incident shell triangle. Shared-face data are checked on both shells. Rejected proposals remain recorded identity operations.

Proposition 5.1 (Guarded winding and uniform isolated band)**.** *Every admitted finite history preserves the two shell degrees. Its spatial Hamiltonian has exactly four isolated modes with* $|E| \le r = b^2/(M-4)$ *and complementary spectrum outside* $(-g, g)$, $g = M-2$, *where* $b^2 = 7/4$ *for adjacent unit shells and* $b^2 = 3/2$ *for separated shells. For two admitted band projectors* Π_v, Π_w,

$$\|\Pi_w - \Pi_v\| \le \beta = \frac{4M\eta c}{d^2}, \qquad c = \frac{g+r}{2}, \quad d = \frac{g-r}{2}.$$

The statements are uniform in the exterior volume.

Proof. During an accepted replacement, every interpolated shell vector is a convex combination of nonzero vectors with pairwise nonnegative scalar products. Its squared norm is at least $\sum_i t_i^2 \|v_i\|^2 > 0$, so normalization gives a shell homotopy. Every noncore onsite eigenvalue stays $\pm M$ and the four core coordinates stay zero; the compact-source Schur proof therefore gives the same band count and localization bounds. Normalization gives $\|H_w - H_v\| \le 4M\eta$. On $|z| = c$ both resolvents have norm at most $1/d$. Their identity and the contour formula for the two projectors give the displayed bound. □

For $\beta < 1$, define

$$S = \Pi_w \Pi_v + (1 - \Pi_w)(1 - \Pi_v), \qquad U = S(S^*S)^{-1/2}, \qquad S^*S = 1 - (\Pi_w - \Pi_v)^2.$$

Since $S\Pi_v = \Pi_w S$ and the positive factor commutes with Π_v, U is unitary and $U\Pi_v U^* = \Pi_w$. The literal particle-hole relation $H_v C = -C\overline{H_v}$ implies $\Pi_v C = C\overline{\Pi_v}$, and the polar formula gives $UC = C\overline{U}$, hence $UCU^T = C$. Thus $U \otimes I_{45}$ transports all twelve uniform native currents and the same pairing tensor $C \otimes \overline{K}$. For $W' = UW$, the pairing matrix $W'^* C \overline{W'}$ is unchanged, while $A_w = W'^* H_w W'$ retains the actual changed spatial energies. Second quantization also transports $N_v = d\Gamma(\Pi_v \otimes I_{45})$ and $\lambda(N_v - 90)^2$.

The explicit source uses $M = 64$, $\eta = 1/512$ and four equiprobable proposals at each record: two opposite transverse changes at the shared-face center, one forbidden zero and one forbidden core change. Its three texture states have transition rows $(1/2, 1/4, 1/4)$, $(0, 3/4, 1/4)$ and $(0, 1/4, 3/4)$. All twelve state/proposal records are retained. Each branch applies its texture transition and $\sqrt{q_r}\,\Gamma(U_r)$, with identity control for a repeated texture, so the complete instrument preserves trace. The band is transported exactly at record endpoints. Omitting U_r instead gives leakage probability at most $\min(1, n^2\beta^2)$ after n quenches and intervening source evolution; the numerical controls exhibit nonzero leakage.

A nearest-neighbor snake through the $N = 2(2L+1)^3$ spatial-spin coordinates compiles each U by adjacent Givens rotations: at most $N(N-1)/2$ two-mode gates and N onsite phases. The complete $N = 686$ controls replay 234,955 two-mode gates per nontrivial texture pair. Addressed quadratic pulses, their timing and drift cancellation are supplied services; this is not a compilation into unchanged C/E repair. Intermediate pulses need not preserve the band, and the controller supplies the endpoint energy change, bounded by $2r$ on the one-body band. Erasing the guards permits zero-degree textures with the same four-mode count. The result establishes controlled protection and transport on this law, without autonomous core motion, local Gauss dressing, physical species identification or relativistic dispersion. The complete proof, rational bounds and gate replay are in `model/compact_defect_transport/`.

5.5 A quartic interaction on the complete compact band

The spatial source is the cubic operator $h = T + Mn \cdot \sigma$, $\|T\| < 2$, $M \ge 8$. Its direction is the normalization of $(x, y, z^2 - a^2)$ inside the cube of radius $a + 1$ and a constant outside, with zero amplitude at the two cores $(0, 0, \pm a)$. The four-band bound is $r = b^2/(M - 4)$, where $b^2 = 7/4$ for $a = 1$ and $b^2 = 3/2$ otherwise. The spectral and particle–hole identities are proved for this source in `model/compact_defect_pair/PROOF.md`. Let W contain the four localized modes and set $A = W^\dagger h W$. The isometry $E = W \otimes I_{45}$ embeds 180 annihilators into the complete finite field. The current and pairing intertwiners hold for every complete memory record, with A retained in the quadratic core. Write $P = EE^\dagger$ and choose

$$V = \lambda\big(N_W - 90\big)^2, \qquad N_W = d\Gamma(P), \qquad \lambda \ge 0. \tag{12}$$

This is an explicit controller design. The particle–hole identity $C\overline{W} = WS$ ensures that the pairing has no localized/complementary cross term. All terms are even, so $H_r^{\mathrm{full}} = H_{r,W} \otimes I + I \otimes H_{r,\perp}$ on every record. The interaction is parity even and commutes with all twelve native internal currents. In the localized occupation basis it contains the quartic term $2\lambda \sum_{i<j} n_i n_j$. The pairing Hamiltonian does not conserve N_W, and its commutator with V need not vanish. Every record preserves the same localized algebra because V acts entirely on that factor.

Hold the texture fixed throughout the memory averaging interval. The complete record limit has a bound independent of exterior volume. Writing the compact energies as $\{-e_2, -e_1, e_1, e_2\}$ with $e_i \le r$, the kinetic norm is $45(e_1 + e_2) \le 90r$. If g is the source feedback gain, the pairing bound and $0 \le N_W \le 180$ give

$$\|H_{r,W}\| \le M_W = 90r + 90(1 + g) + 8100\lambda.$$

The radial two-column zero-mode construction is a separate special case with ninety annihilators, centre 45 and bound $45(1+g)+2025\lambda$. The actual memory transition has the reset decomposition $T = (1-p)R + pS$, and contracts zero-sum operator-valued memory blocks by at most p. At step size t/n, its reduced core state consequently satisfies

$$\|\rho_{W,n}(t) - e^{-it\bar{H}_W}\rho_W(0)e^{it\bar{H}_W}\|_1 \le \frac{4M_W t}{n(1-p)} + \left(\frac{8}{1-p} + 4\right)\frac{M_W^2 t^2}{n}. \tag{13}$$

The estimate includes arbitrary initial classical–quantum memory correlations and a finite untouched reference. The first term vanishes for stationary product memory. The mean Hamiltonian contains the same memory-derived mass matrix and the interaction (12). Uniformity concerns this fixed localized algebra under its source isometries; it gives no uniform continuum estimate for all spatial observables.

Equation (13) compares the unread core. There is a separate estimate for the full native history register. For a fixed initial memory label, purify the core/reference state. Write p_ω for the field-independent history probability, $\psi_\omega = U_\omega \psi$, and $\bar{\psi} = e^{-it\bar{H}_W}\psi$. If δ is the unread trace-norm bound, then $\sum_\omega p_\omega(1 - |\langle\bar{\psi}|\psi_\omega\rangle|^2) \le \delta/2$. The pure-state distance identity and Jensen's inequality bound the full history/core trace distance from $\sum_\omega p_\omega |\omega\rangle\langle\omega| \otimes |\bar{\psi}\rangle\langle\bar{\psi}|$ by $\sqrt{2\delta}$. Partial trace covers mixed inputs. For initial classical memory/core correlations, condition on the initial memory label and retain that tag in the ideal history law; convexity preserves the same bound. The complete final record/core rate is therefore $O(n^{-1/2})$. Opposite qubit phase records saturate this square-root relation. No uniform estimate for rare conditioned histories or intervening field measurements follows.

The interaction is defined through a spectral projector and need not be strictly onsite. For the radial special case, the resolvent estimate at texture coefficient 64 and $W_x^\dagger W_x = \rho_x I_2$ gives, for the projection A_L onto $|x|_\infty < L$,

$$\|(I - A_L)E\|^2 \le \epsilon_L = \frac{3}{2 \cdot 62^2}\, 32^{-2(L-1)}.$$

Thus $\|P - A_L P A_L\| \le 2\sqrt{\epsilon_L}$, and finite rank gives $\|N_W - d\Gamma(A_L P A_L)\| \le 360\sqrt{\epsilon_L}$. The radial interaction $\lambda(N_W - 45)^2$ is exponentially localized about the prescribed origin, without being a strictly onsite interaction.

On this radial subspace there is an exact obstruction for the onsite density alternative. On the two-particle sector let $D_g = \sum_x g_x \bigwedge^2 P_x$ and $E_2 = \bigwedge^2 E$. Direct compression gives

$$E_2^\dagger D_g (I - E_2 E_2^\dagger) D_g E_2 = \left[\sum_x g_x^2 \rho_x^2 - \left(\sum_x g_x \rho_x^2\right)^2\right] I. \tag{14}$$

Writing $c = \sum_x \rho_x^2 < 1$, Cauchy–Schwarz bounds the bracket below by $(1-c)\sum_x g_x^2 \rho_x^2$. Every nonzero member of this onsite density class leaks from the source subspace on which it has support. This statement does not exclude a dressed low-energy subspace or other local interaction tensors.

Connected dynamics and spectral shifts. The compact mean pairing operator has 84 charged two-mode blocks and twelve unpaired neutrino modes. Select the two spatial pairs of one charged-lepton singular value m. Fix one occupation in each of the other 82 blocks and fill six neutral modes. Pair creation and annihilation kill these spectators, whose number is 88. Subtracting their fixed energy gives the exact reducing four-mode Hamiltonian

$$H_4 = e_1(n_1 - n_2) + e_2(n_3 - n_4) + m(a_1^\dagger a_2^\dagger + a_2 a_1 + a_3^\dagger a_4^\dagger + a_4 a_3) + \lambda(N_4 - 2)^2.$$

This is an invariant sector of the mean Hamiltonian, not its asserted vacuum or an invariant sector of every finite memory pulse. Pair parities split its sixteen levels into the four even/even values $0, 4\lambda, 2\lambda \pm 2\sqrt{\lambda^2 + m^2}$, four odd/odd values $\pm e_1 \pm e_2$, and eight mixed values $\lambda \pm e_i \pm m$. Consequently

$$E_0 = \min\{2\lambda - 2\sqrt{\lambda^2 + m^2}, -e_1 - e_2, \lambda - m - \max(e_1, e_2)\}.$$

This enumeration is a complete diagonalization by pair parity. For the unique odd/odd ground, a positive-energy orbital has two addition-plus-removal poles $e_1 + \lambda \pm m$, each of residue $1/2$. At $M = 64$, half-separation one and $\lambda = 1/128$, rational coefficient and exterior-volume enclosures place all three inherited charged-lepton slices in this ground sector:

Quadratic coefficient m	Lower pole	Upper pole
0.000002072030404	0.01171632730	0.01172047136
0.0004284302055	0.01128996912	0.01214682953
0.007205207119	0.004513192209	0.01892360645

The unit is the declared source pulse energy. These are slice poles, without a physical mass assignment or a determination of the full interacting vacuum. The compact exterior error δ_L bounds each displayed pole error by δ_L in this certified sector; no complementary mode is discarded.

The full 180-mode ground also has an explicit variational bracket. List the charged singular values, including colour multiplicities, as m_j, $j = 1, \ldots, 21$. Each occurs in two pairing blocks at each spatial energy e_i, with quadratic levels $\{\pm m_j, \pm e_i\}$. The twelve neutral modes contribute $-3(e_1 + e_2)$, giving

$$E_{\mathrm{quad}} = -2\sum_{j=1}^{21}[\max(m_j, e_1) + \max(m_j, e_2)] - 3(e_1 + e_2).$$

Positivity of V makes this a lower bound. A quadratic product ground has mean occupation 90 and number variance one per even pairing block, zero per odd block. If b blocks are even, its trial energy is $E_{\mathrm{quad}} + \lambda b$. A completely blocked trial state has exact occupation 90, zero interaction energy and kinetic energy $-45(e_1 + e_2)$. Thus, retaining every localized mode,

$$E_{\mathrm{quad}} \le E_{\mathrm{full}} \le \min\{E_{\mathrm{quad}} + \lambda b, -45(e_1 + e_2)\}.$$

The specified coefficient witness has $b = 28$. These bounds constrain the full vacuum energy without identifying its excitation spectrum.

The radial control sets $e_1 = e_2 = 0$. Its ground energy is $2\lambda - 2\sqrt{\lambda^2 + m^2}$, and its single-mode spectral poles and weights are

$$\omega_\pm = 2\sqrt{\lambda^2 + m^2} - \lambda \pm m, \qquad w_\pm = \frac{1 \mp m/\sqrt{\lambda^2 + m^2}}{2}.$$

In this radial control the cross-pair density cumulant is $-\lambda/(4\sqrt{\lambda^2 + m^2})$, while its normal and anomalous cross-pair two-point contractions vanish. Hence positive λ gives a non-Gaussian connected interaction. These spectral shifts preclude identifying the quadratic mass coefficients with the interacting excitation energies without further calculation.

A common finite energy and modular response. For the same full localized Hamiltonian choose $\rho_\beta = Z^{-1} e^{-\beta \bar{H}_W}$, $\beta > 0$. This state is faithful. A global spectral reset with Kraus operators $\sqrt{p_i}|i\rangle\langle j|$ prepares it and preserves total probability. That reset is a declared service, not an implementation by the original site repair. For $H(s) = \bar{H}_W + sX$, Duhamel differentiation gives

$$\rho'_\beta(0) = -\beta \int_0^1 \rho_\beta^u (X - \langle X \rangle) \rho_\beta^{1-u}\, du.$$

The positive logarithmic-mean kernel yields $F' = \langle X \rangle$, $F'' = -\beta C_{\rho_\beta}(X, X) \le 0$, and $S' = \beta(E' - \langle X \rangle)$, where C_ρ is the Bogoliubov–Kubo–Mori covariance. The noncommuting choice $X = (N_4 - 2)^2$ is checked against an independent matrix-exponential derivative. These are finite identities for one nonquadratic Hamiltonian. They supply no geometric modular flow, local relativistic stress, area normalization or geometric entropy stationarity, and therefore imply no Einstein dynamics. The exact characteristic polynomial, complete finite spectra, record control and source-bound replay are implemented in `model/compact_interacting_response`; the radial controls and onsite leakage calculation are in `model/interacting_defect_response`.

5.6 Work on the moving compact interacting band

A deformation must act on the same Hamiltonian used to identify the matter states. On the compact opposite-degree texture, vary the actual hopping by $h_s = sT + MV$, with $1/2 \le s \le 3/2$, $M \ge 8$ and $\|T\| < 2$. The texture, memory coefficients and quartic coupling are fixed. The four-dimensional spectral band has energies in $[-3, 3]$ and separation at least $M - 6$ from its complete complement. Let P_s be its projector, C the source particle-hole matrix, and K_ω the internal pairing kernel on a complete memory record ω. The same source law is

$$N_s = \mathrm{d}\Gamma(P_s \otimes I_{45}), \qquad H_{\omega,s} = \mathrm{d}\Gamma(h_s \otimes I_{45}) + \mathrm{pair}(\mathsf{C} \otimes \overline{K_\omega}) + \lambda(N_s - 90)^2.$$

Its exact force conjugate to the hopping control is

$$\partial_s H_{\omega,s} = \mathrm{d}\Gamma(T \otimes I_{45}) + \lambda\{N_s - 90, \mathrm{d}\Gamma(P'_s \otimes I_{45})\}. \tag{15}$$

The second term follows by differentiating the noncommuting square. Freezing the localization projector omits this contribution to source work.

Proposition 5.2 (Localized force with the moving band retained)**.** *The band admits a unitary comparison frame W_s in which its pairing form is constant and its intrinsic force is*

$$J_s = \mathrm{d}\Gamma((W_s^* T W_s) \otimes I_{45}), \qquad \|J_s\| \le 360, \qquad \|P'_s\| \le \frac{6}{M - 6}.$$

For two even localized states with the same even complement reference, their contrast of (15) *equals their contrast of J_s. The bounds are independent of the containing finite cube.*

Proof. For band vectors p and complementary eigenvectors q, $\langle q, P_s' p\rangle = \langle q, Tp\rangle/(E_p - E_q)$ and the diagonal blocks of P_s' vanish. Hence $\|P_s'\|^2 \le \|P_s'\|_{\mathrm{HS}}^2 \le 8\|T\|^2/(M-6)^2 < 36/(M-6)^2$. Set $L_s = [P_s', P_s]$ and solve $U_s' = L_s U_s$. Then $P_s = U_s P_1 U_s^*$ and $W_s = U_s W_1$. Particle-hole covariance gives $\mathsf{C}\overline{U_s} = U_s\mathsf{C}$, so the pairing form is fixed. Differentiating $A_s = W_s^* h_s W_s$ gives $A_s' = W_s^*(T + [h_s, L_s])W_s = W_s^* T W_s$: h_s is block diagonal and L_s off diagonal. Number and pairing therefore have no derivative in this transported frame. The 180-mode second-quantization norm is at most $180\|T\| < 360$. In an even product state the cross-band terms have zero expectation; subtracting the same complement reference cancels its remaining energy. □

At fixed s, a complete-memory trace-norm error δ consequently gives a force-readout error at most 360δ. The comparison transporter is an explicit mathematical identification; Hamiltonian evolution with changing s does not implement it automatically. For the supplied Gibbs reference of the same mean Hamiltonian at fixed $\beta > 0$, Duhamel differentiation gives

$$F'_{\mathrm{thermal}} = \mathrm{Tr}(\rho_s J_s), \qquad S' = \beta\big(E' - \mathrm{Tr}(\rho_s J_s)\big).$$

The complete compact source and a reducing sixteen-state interacting slice check these derivatives, retaining the spatial splitting and work performed by the hopping control; `model/compact_band_response/` gives the proof and replay. The hopping amplitude is a declared control. Its work derivative supplies no identification with a metric variation or conserved Lorentz stress, and this finite Gibbs identity supplies no regional geometric modular flow, area normalization or fixed-volume generalized-entropy stationarity.

5.7 A finite local Gauss law for the native currents

The native fifteen-state matter representation decomposes into invariant blocks V_r of dimensions $6, 3, 3, 2, 1$. A declared finite link design preserves those blocks. For an oriented edge $x \to y$, set

$$\mathcal{L}_{xy} = \mathbb{C}|0\rangle \oplus \bigoplus_r (V_{r,x}^* \otimes V_{r,y}), \qquad U_{ab} = |\bar{a}, b\rangle\langle 0|,$$

where $U_{ab} = 0$ when a, b belong to different blocks. The link dimension is $1+36+9+9+4+1 = 60$. For any native anti-Hermitian generator T, its endpoint generators are $0 \oplus \bigoplus_r (-T_r^T \otimes I)$ and $0 \oplus \bigoplus_r (I \otimes T_r)$. Adding the local fermion charge $d\Gamma_x(T)$ defines $G_x(T)$ on the complete Fock and link space.

For real J, the same finite Hamiltonian transports matter and compensating link charge:

$$H_{xy} = J\sum_{a,b} c_{xa}^\dagger U_{ab} c_{yb} + \mathrm{h.c.} + \epsilon(I - |0\rangle\langle 0|), \qquad [G_v(T), H_{xy}] = 0, \quad \epsilon \ge 0.$$

Indeed, the two link commutators are $[L_x(T), U_{ab}] = -\sum_i T_{ai} U_{ib}$ and $[L_y(T), U_{ab}] = \sum_j T_{jb} U_{aj}$. They cancel the creation and annihilation commutators of $d\Gamma(T)$ after relabeling the summed indices. The link-energy projector commutes with both generators. This proves the identity for all occupation sectors and finite sums of edges. It also proves preservation of the Gauss kernel by evolution and by complete instruments whose Kraus operators commute with the generators.

For an explicit nontrivial sector, supply a static dual reference at y. Within one block of dimension d, the vectors

$$A = \sum_b |y, b\rangle|\bar{b}\rangle|0\rangle, \qquad B = \sum_{a,b} |x, a\rangle|\bar{b}\rangle|\bar{a}, b\rangle$$

are Gauss singlets, with norms $\sqrt{d}$ and d. The Hamiltonian sends A to JB and B to $dJA+\epsilon B$. Its orthonormal two-state restriction is $\left(\begin{smallmatrix} 0 & J\sqrt{d} \\ J\sqrt{d} & \epsilon \end{smallmatrix}\right)$. All twelve native generators are checked on every block. Exact Cayley steps and complete endpoint readouts retain all eighty histories obtained from a five-outcome initial block measurement and four binary observations. The change of this same Hamiltonian's energy under each observation is included as detector work; the detector reservoir is a supplied service.

This construction supplies finite non-Abelian operator Gauss constraints. The link operators are not unitary Wilson links. The link register, static reference, coherent hopping and readout are chosen inputs, with no asserted compilation into bare C/E repair. The localized compact band has a separate state inclusion; dressing it into this local Gauss sector requires an additional construction. Neither the finite transport frequency nor current conservation establishes physical particle masses, a chiral continuum or anomaly cancellation of its regulator. The exact source and independent full-Fock control are in `model/native_quantum_gauss`.

6 The conditional catalogue and the physical comparison

6.1 What the displayed source actually assigns

The common finite construction specifies three distinct charged-lepton coefficients, six quark coefficients with color multiplicities, and three zero neutrino coefficients in a 45-component internal kernel. The gauge and Higgs directions belong to the separate conditional bosonic action. The complete assignment is therefore:

Channel	Displayed source input	Established finite meaning
Charged leptons	$Y_e = \sqrt{2}\, g Z_C$	Three memory-derived singular coefficients on the native charged-lepton slots
Up and down flavors	Diagonal powers of 1/6	Six declared coefficients on native color slots; this displayed choice has identity quark mixing
Minimal neutrino slots	Zero	Three zero internal coefficients, without a massless relativistic particle proof
Gauge and scalar fields	Canonical one-doublet action, g_i, λ_H, v	Conditional symmetry breaking and quadratic masses; no bosonic defect-state inclusion in the CAR construction

This is a complete accounting of the displayed model's assignments, not a complete catalogue of nature. The quark exponents, scalar quartic, neutrino operator and physical flavor mixing are not determined by the charged-lepton memory inverse. The finite gauge theorem fixes an allowed Lie algebra and, under the matter attachment, internal charges; it leaves these dynamical coefficients unconstrained.

Two mathematical distinctions prevent a false physical spectrum. First, the compact four-mode band has a nonzero operator $A = W^\dagger h W$. Its energies enter the exact common operator and cannot be removed to retain the zero-mode mass list. Second, the positive quartic coupling changes the spectral poles. The pixel feedback calculation uses quadratic coefficients, rather than

the changed interacting current spectral measure. The compact interacting operator and its exact slice readouts therefore require a recalculated electromagnetic response before a common interacting feedback fixed point can be claimed.

The concrete physical target is the joint map

$$\begin{aligned}&\text{(same source, vacuum, clocks, instruments)}\\ &\quad\longrightarrow \text{(total defect energies, current residues, dispersion, vertices)}\\ &\quad\longrightarrow \text{(one response kernel and its pixel fixed point).}\end{aligned}$$

Its first row must supply an autonomous finite-energy defect process, rather than an externally fixed texture. Its second must retain all spatial regulator modes until an actual decoupling or continuum theorem controls them. The third must use the first two, including thresholds and physical normalization. These are composition requirements on one permitted design; uniqueness among all possible simulator designs is not a prerequisite.

6.2 Particle observables and comparison conventions

At finite volume the spectral measure of a prepared operator O is

$$d\mu_O(\omega) = \sum_n |\langle n|O|\Omega\rangle|^2 \delta(\omega - (E_n - E_\Omega))\, d\omega.$$

A visible eigenvalue requires nonzero residue in the actual detector channel. The physical dictionary must transport O, the state, the Hamiltonian, current charges and the clock together. Localized wavefunctions have position and momentum distributions; they are not momentum eigenstates, and a Fourier transform of a pinned mode is not a moving-defect dispersion experiment.

In a justified relativistic regime a stable massive excitation has a common rest-energy and propagation relation. Setting the numerical speed of light fixes units only. The same source must establish the relative normalization of time and space, the momentum generator and the dispersion law. An unstable particle requires a resonance convention and width [11]. Charged sectors coupled to a massless photon require an infrared treatment rather than an assumed isolated sharp particle pole [12, 13]. A massless photon also precludes a strictly positive global infinite-volume vacuum gap; massive-sector thresholds are separate statements. Color confinement prevents reading a free color-mode frequency as an observable quark or gluon mass.

Comparison uses mass ratios in one stated running or pole convention, physical mixing observables, widths and interaction strengths. Running charged masses and mixing data must share scale and scheme [9, 10]. Quark and unstable-particle conventions cannot be replaced by arbitrary ratios of finite-cavity levels. Parameters chosen after exposure to these quantities carry that provenance; independent tests use data or consequences not used in the choice. The finite source spectra reported here have no certified running-to-pole, confinement, or laboratory-clock map.

6.3 Interaction, chiral continuum and responsive geometry

The quartic result gives a genuine connected interaction on a fixed localized algebra, with a bound uniform in the surrounding radial volume. Its spatial tails and fixed number of localized modes do not by themselves define an interacting local 3+1 field theory. A controlled common field limit must also establish the intended chiral measure, anomaly realization, local gauge constraints, surviving interactions and regulator census. Finite internal anomaly sums are necessary representation checks; a chiral lattice measure has additional locality and integrability requirements [14, 15, 16].

The same matter energy and stress must then enter the geometric response. The companion Einstein implication uses compatible clocks and metric, conserved stress, a geometric modular normalization and controlled fixed-proper-volume generalized-entropy variations [4]. The finite Gibbs first law proved for the interacting localized Hamiltonian is a useful common-operator identity. It does not supply a geometric area term, the small-ball remainder, the Ward/Bianchi map or a universal Newton normalization. Einstein dynamics cannot be inferred by joining unrelated matter and entropy calculations.

Reproduction and mathematical scope

The executable sources and exact certificates accompany the public repository [1]. The principal components used in this construction are `model/conditional_standard_model`, `model/mcpr_native_memory_bridge`, `model/pixel_mass_closure`, `model/defect_current_bridge`, `model/compact_defect_pair`, `model/compact_defect_transport`, `model/compact_interacting_response`, `model/compact_band_response`, and `model/native_quantum_gauss`. They distinguish exact algebra and rational interval proofs from floating spectral checks and empirical comparisons. A source hash identifies a calculation; its mathematical scope is the stated theorem and domain.

The finite results establish a common current/mass operator, complete-record memory limits, a conditional feedback closure, a compact finite-energy background with isolated modes, guarded band transport, a common localized interaction and its hopping-work response. A separate native link design supplies exact local Gauss constraints. A source-derived physical spectrum additionally requires their common interacting realization with a physical state, clock and response dictionary.

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